

Digital Control of Four Switch Synchronous Buck-Boost Power Converter based on PIC18F4520 Microcontroller

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Abstract—The paper presents a practical realization of digital voltage control of a synchronous "buck-boost" power converter, with a wide range of input voltage changes. The power converter is adapted to the power from input solar panels and for different types of output loads. The paper analyzes the basic principles of above mentioned power converter in continuous conduction mode. Also, the paper presents a control strategy based on the PIC18F4520-MICROCHIP microcontroller. All measuring signals (input and output voltage, as well as output current) are connected to the input of the digital controller via galvanically isolated measuring circuits, with appropriate A/D conversion. Finally, some of the most important experimental results obtained during testing of the synchronous „buck-boost“ power converter at different input voltages and different output loads are presented.

Keywords- Digital control; synchronous operation; buck-boost; A/D converter; voltage correction; PIC microcontroller

I. INTRODUCTION

The use of conventional DC-DC voltage converters (step-up/step-down) that contain, in addition to the controllable switching elements (MOSFETs or IGBTs) the fast diodes, is associated with the problem of power losses and consequently that, with reducing of efficiency. This is especially pronounced at relatively low input voltages and in cases where, due to the requirement for relatively high output power, relatively large currents are present[1-3].

This case is characteristic for PV "off-grid" systems that use battery storage elements, whose voltages are relatively low (typically 12Vdc and 24Vdc) [2], [4].

In the balance of losses, the largest share belongs to losses in the conduction state (on-state) and commutation losses on switching elements. To overcome this shortcoming, various topologies of synchronous DC-DC power converters have been developed [3], [5], in which the basic switching elements are MOSFET switches with a low value of drain-source resistance in conduction state $R_{ds(on)}$ of only a few m Ω . This application

excludes the use of fast diodes and their role is taken over by MOSFET switches.

One topology of a synchronous DC-DC power converter based on four fast switches, the so-called "buck-boost four switch power converter", very interesting for practical applications in "off-grid" solar systems is the subject of consideration in this paper. The basic topology of a synchronous "buck-boost" power converter based on the use of four fast switching elements (MOSFET switches M1-M4) is shown in Fig.1.

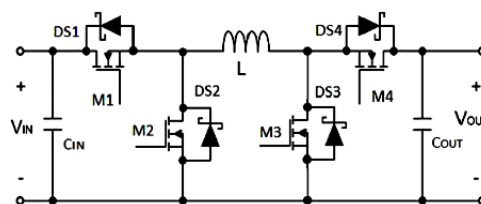


Figure 1. The basic topology of "buck-boost four switch converter"

The power converter consists of two half-bridges, where one is supplied from the input side with the voltage V_{in} , while the other is supplied from the output side with the voltage V_{out} . In the operating mode of the power converter, switches M1 and M3 are switched on simultaneously at the time in interval t_{on} , while switches M2 and M4 are switched on simultaneously in te time interval t_{off} .

The accumulation elements of importance in this power converter are the output capacitor C_{out} and the inductance L . Depending on the duration of the driving pulses of the switching transistors, energy is transferred from the coil to the capacitor and the output voltage is the same polarity as the input, but may be less than the value of the input voltage ("buck" converter) or greater than the input voltage ("boost" converter). References [3], [5-6] show the basic principles of operation of this topology of power converter, as well as some of its possible applications. This paper presents a possible

implementation of digital output voltage control for this type of power converter, based on a PIC18F4520 microcontroller.

The paper is organized as follows: Chapter II presents an analysis of the operation of the power converter and the scope of the characteristic waveforms. Chapter III presents the basic digital control structure of the power converter and a description of the key control functions. Chapter IV describes the software implementation. Chapter V gives an practically realization of power converter, with a description of individual functional blocks, and finally in Chapter VI, the key experimental results are presented.

II. THE ANALYSIS OF POWER CONVERTER OPERATION

The analysis and basic principles of operation of the "buck-boost" converter from Fig.1 are presented in this chapter. Depending on the value of the „duty cycle“ coefficient D ($0 < D < 1$), it is determined in which mode the power converter will operate ("buck" or "boost").

The coefficient $D = t_{on}/T_s$, where t_{on} is the time interval during which the transistors M1 and M3 are switched on, while T_s is the value of the switching period (reciprocal value of the switching frequency f_s , i.e. $T_s = 1 / f_s$).

Fig.2 shows the equivalent electrical circuit of power converter for the time interval $0 < t < DT_s$.

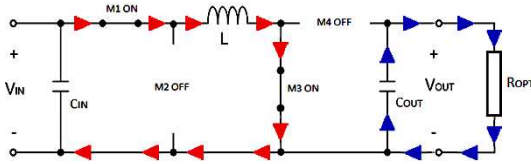


Figure 2. The scope of equivalent electrical circuit of power converter for time interval $0 < t < DT_s$

Switching transistors M1 and M3 are switched on in this interval, and M2 and M4 are switched off. The current through the inductor L increases, whereby this inductance is charged by magnetic energy. During this interval, the load is supplied only with energy from the output capacitor C_{out} , and the output voltage V_{out} decreases.

Fig.3 shows the equivalent electrical circuit of power converter for the time interval $DT_s < t < T_s$. During this interval, switching transistors M1 and M3 are switched off, and switching transistors M2 and M4 are switching on.

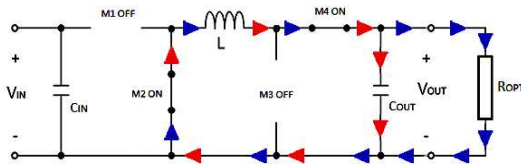


Figure 3. The scope of equivalent electrical circuit of power converter for time interval $DT_s < t < T_s$

During this interval, the load receives energy from the inductance coil. The inductance in this case acts as a current source that charges the capacitor and thus leads to an increase

in the voltage across the capacitor C_{out} . Capacitor C_{out} "absorbs" the variable component („ripple“) of the coil current, while the load takes over the DC component of this current.

The characteristic waveforms for the presented topology of the power converter and for the case of continuous conduction mode (CCM) are shown in Fig.4.

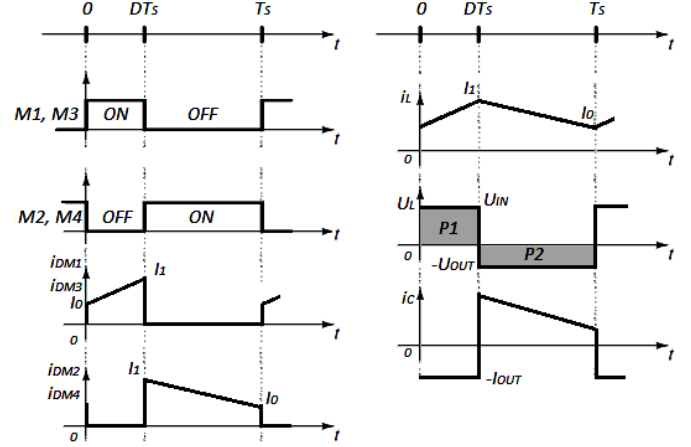


Figure 4. The characteristic waveforms of power converter from Fig.1, for continuous conduction mode (CCM).

Since the average value of the voltage at the inductance L is equal to zero, we can write a relation that refers to the voltage integral, i.e. to the volt-seconds " $V \cdot s$ " (or to the characteristic areas $P1$ and $P2$):

$$P_1 + P_2 = 0 \quad (1)$$

$$P_1 = V_{in} \cdot DT_s \quad (2)$$

$$P_2 = -V_{out} \cdot (1-D)T_s \quad (3)$$

After replacing expressions (2) and (3) in equation (1), we obtain the following relations:

$$V_{in} \cdot DT_s - V_{out} \cdot (1-D)T_s = 0 \quad (4)$$

$$V_{in} \cdot DT_s = V_{out} \cdot (1-D)T_s \quad (5)$$

Finally, a definite expression is obtained for the voltage transferr ratio of the "buck-boost" power converter CCM mode:

$$V_{out} = V_{in} \cdot \frac{D}{1-D} \quad (6)$$

For the range of the coefficient D , $0 < D < 0.5$, the power converter operates in the "buck" mode. For the range of the coefficient D , $0.5 < D < 1$, the power converter operates in the "boost" mode. For the value $D = 0.5$, the power converter operate in the mode when the input and output voltages are equal, i.e. $V_{in} = V_{out}$.

III. DESCRIPTION OF BASIC CONTROL STRUCTURE OF POWER CONVERTER

The basic control scheme of the power converter is given in Fig.5. The control part is based on a digital micro-controller PIC18F4520, manufactured by MICROCHIP, which performs voltage measurements via an internal A/D converter and based

on these measurements controls the operation of the "buck-boost" converter. Optionally, the measurement results can be shown on the display, which consists of four seven-segment indicators.

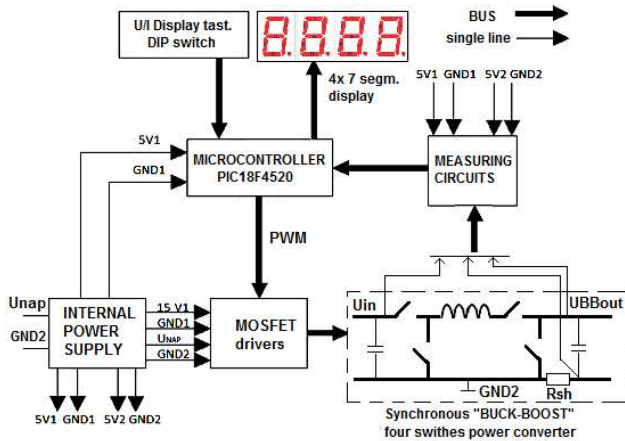


Figure 5. Basic control structure of four switch "buck-boost" power converter

The microcontroller measures three voltages and one current: (1) voltage at the input of the power converter V_{in} (which is variable in a wide range 8-40Vdc), (2) voltage at the output of the power converter V_{out} , (3) supply voltage of the internal control electronics $V_{nap} = 12V$ (range 9-18V) and (4) load current over the measuring shunt R_{sh} .

Based on the performed measurements, the microcontroller adjusts (corrects) the voltage at the output of the power converter indirectly by increasing or decreasing the pulse duration at its PWM outputs (PWMA and PWMB to excite the corresponding drive circuits i.e. transistors M1-M3 and M2-M4, respectively). The PIC18F4520 has PWM outputs that are designed to control the configurations of half-bridge or full-bridge power converters. Also, the microcontroller monitors the status of the switches and buttons and, if necessary, the user updates the set voltage at the output of the converter.

The internal power supply required for the microcontroller and other digital circuits is galvanically separated from the power supply of the device and from the voltage at the input of the power converter. Within the internal power supply circuit, there is a DC-DC converter that produces galvanically separated voltages of +15V and -15V from the U_{nap} input voltage. From the voltage +15V, through the voltage stabilizer LM7805, the voltage +5V (5V1) required for powering the microcontroller and other electronic circuits is obtained. From the U_{nap} voltage through the additional voltage stabilizer LM7805, the voltage +5V (5V2) required for the supply of isolating operational amplifiers used to measure the mentioned voltages and shunt current was obtained.

The control circuit uses IR2110 drivers for MOSFET switches with "bootstrap" coupling. This coupling generates the voltage required to quickly turn-on the high-side MOSFET transistors from a „floating“ power supply. Via the "DIP-switch" element it is possible to adjust the voltage (voltage reference) at the output of the converter.

IV. THE PRACTICAL REALIZATION OF POWER CONVERTER

After a detailed design of the power converter (calculations and simulations), its practical realization was approached. Fig.6 shows the electrical scheme of the realized synchronous "buck-boost" power converter based on the use of four switching elements (MOSFET switches M1-M4).

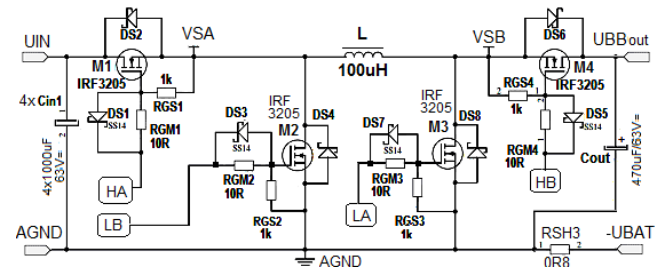


Figure 6. Electrical scheme of realized four switch "buck-boost" power converter

The project envisages high-current HEXFET MOSFET switches IRF3205 with ultra-low resistance $R_{ds(on)} = 8m\Omega$ and for a continuous current of 80A. The maximum voltage drain-source can withstand is $V_{ds_max} = 55V$. Despite the fact that each of the MOSFET switches contains an internal diode with the characteristic 1.3V / 80A / 100ns, in parallel for each of them, fast Schottky diodes DS2, DS4, or DS6, DS8, type SS16 with a withstand voltage of 60V are provided.

The power converter is designed to achieve a maximum capacitor voltage ripple of 100mV at a switching frequency $f_s = 50kHz$ and an output current of 4A. For these values and for the adopted $D = 0.6$, it is calculated that a capacitor value of 100µF is required. In order to achieve the lowest possible voltage ripple, the output capacitor $C_{out} = 470\mu F / 63Vdc$ was selected. The projected value of the inductor L is 100µH, for a current of 10A and this design meets the needs in terms of the required output current and current ripple. ("ripple"). The projected values of the resistors RGM1-RGM4 in the gates of the MOSFET switch M1-M4 are 10Ω, while the projected values of the resistor RGS1-RGS4 at the terminals GS of the corresponding MOSFET switches are 1k. Diodes DS1-DS4 serve to provide fast discharge of the capacitance of the GS capacity of the corresponding MOSFET switches.

The measuring of the output current was achieved by a resistive measuring shunt RSH3 which is connected to the (-) pole of the battery (-UBAT). The value of this resistive shunt is 0.8Ω and is designed for a maximum current of 10A.

Fig.7 shows the electrical connection scheme of the PIC18F4520 microcontroller and the associated drive circuits for MOSFET switches. For the realization of the drive circuits of the switches, standard driver circuits IR1, IR2, type IR2110 (International Rectifier) were used. This drivers have two PWM inputs and outputs, for the drive of MOSFET transistors. One output (on „high-side“) is floating while the other (on „low-side“) is standard. In order for this topology to work properly, a "bootstrap" coupling circuit is used. Based on this coupling, the voltage required for fast switching of the floating MOSFET transistor (on „high-side“) is realized.

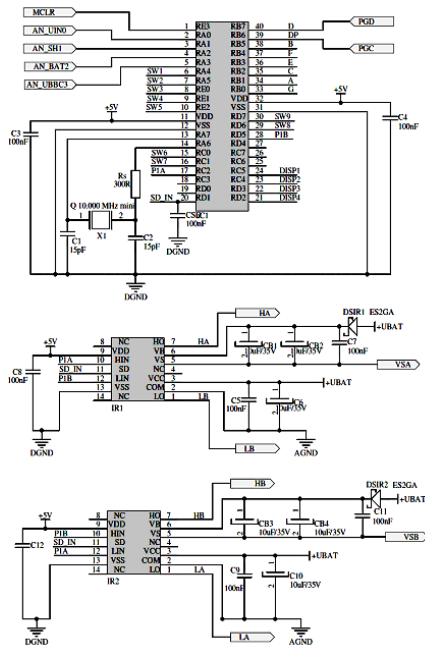


Figure 7. Electrical scheme of PIC microcontroller and driver circuits

The "bootstrap" coupling circuit uses a capacitor that charges when the lower transistor is turned on. In that short time, the capacitor is charged to the V_{cc} value. In this way, it is ensured that the voltage at the gate of „high-side“ MOSFET is for V_{cc} higher than the voltage at the his source. The considered "buck-boost" power converter contains four MOSFET transistors, two of which are always „on-state“ at the same time. This fact is exploited that with two PWM outputs and two drive circuits IR2110, with proper connection, can control the MOSFET switches applied in the considered power converter. In order to prevent both transistors from conducting at the same time, the implemented time function, the so-called "dead-time", during which both PWM signals are inactive.

The electrical connection diagram of the measuring circuits (measuring the input and output voltage of the power converter, as well as measuring the output current) and the comparator circuit for performing the "shut-down" (SD) function via the CP1A comparator is shown in Fig.8.

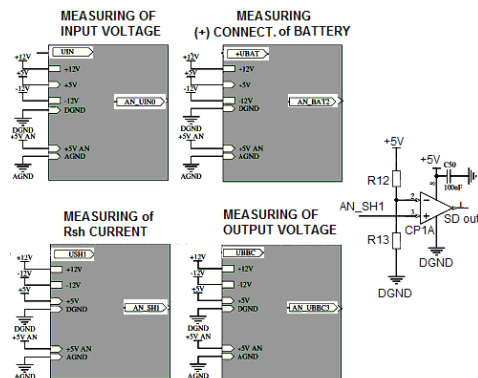


Figure 8. Electrical connection diagram of measuring circuits

The principal measurement scheme for each of the modules (measuring circuits) shown in Fig.8 is given in detail in Fig. 9.

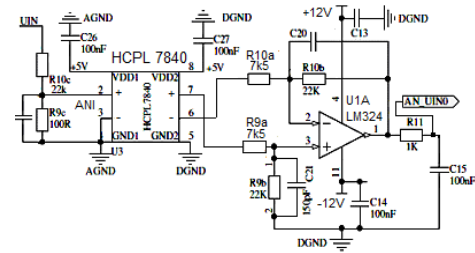


Figure 9. Detailed electrical scheme of each individual measuring circuit

The measuring circuits have basically the same scheme, only the values of the components differ because they are intended for measuring voltage signals of different levels. At the base of all measuring circuits, there is an isolating operational amplifier HCPL7840. The architecture of the HCPL7840 circuit is such that galvanic isolation is provided between the input and output. The input voltage must be in a narrow range, for which the linearity of the circuit is guaranteed at a voltage gain of approximately 8. This circuit is powered by + 5V and needs to be supplied with that voltage on both sides, with these galvanically isolated voltages being referred to the galvanically separated reference points AGND and DGND, as shown in Fig.9. The output of this circuit is connected to the operational amplifier U1A LM324 which amplifies the voltage, which is further led to the input AN_UIN0 of the A/D converter of the PIC1F4520 microcontroller.

V. SOFTWARE IMPLEMENTATION

In Fig. 10, a program flow chart-1 diagram is presented. Within the program, the basic units are singled out, and they are executed when the conditions for that are met.

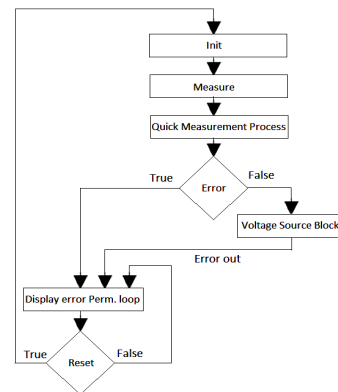


Figure 10. The flow chart -1

After starting the device, the microcontroller is adjusted, and its peripherals are adjusted. After completing the adjustment, the first measurements are made to check the condition of the input voltages. If it is found that any of the input voltages has an irregular value (which is indicated by setting the error indicator), the program code for controlling

the operation of the "buck-boost" converter is not started, but the error status is displayed and the device remains in this state, all until a reset is performed.

If there were no errors when starting the device, the program switches to the "Voltage Source Block" routine. Within this block, input voltage measurements are performed every 2.5 ms. If all measurements are within the permitted range, the value of the set voltage at the power converter output is read from the DIP-switch.

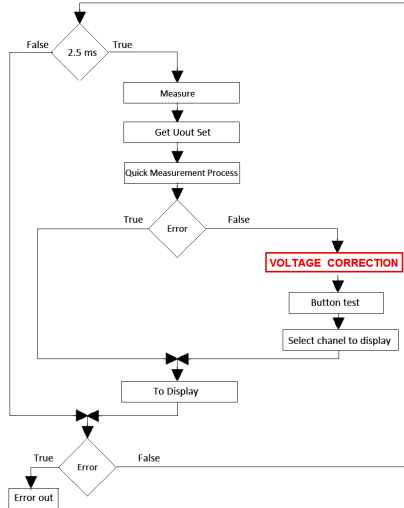


Figure 11. The flow chart 2

Voltage correction is then performed via the "Voltage Correction" function by increasing or decreasing the pulse duration as shown in flow chart 2 on Fig.11. Also, the state of the button is read through which the display of the desired measured voltage is selected. If an error has occurred, the error indicator as well as the error identification indicator are set (the display shows the ordinal number of the error which shows in which part of the code the error occurred). Due to the set error indicator, there is an exit from the infinite loop, and thus from the block "Voltage Source" (see Fig.10).

The "main.h" file is the main program file. It also includes auxiliary files "functions.h" within which are declarations and definitions of functions used in the program and "variables.h", within which are defined variables used in the program. The program code is divided into functions, in order to achieve better visibility of the code. In operating mode, when there are no errors, the microcontroller cyclically executes the code contained in the „Voltage Source FN ()“ function. Within this function, there is an infinite loop „while (1)“ that will execute until an error is detected.

VI. THE EXPERIMENTAL RESULTS

In this chapter, the key experimental results obtained on the laboratory prototype of the circuit of the realized synchronous „four switch buck-boost“ power converter will be presented. The measurements were performed in "buck" and "boost" modes, for a wide range of input voltage changes (in the range of 10-30V) and for resistive load.

A) „Buck“ operation (resistive load)

Fig.12 shows the transient responses (transient load regulation) of the output voltage of the power converter during a sudden change in load current.

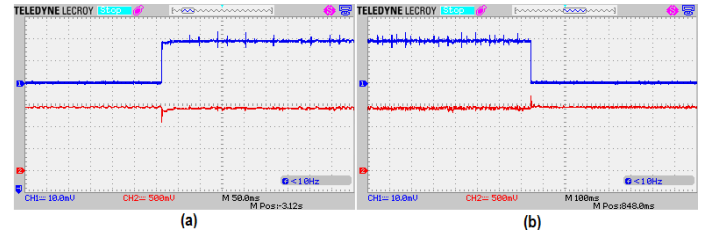


Figure 12. Transient response of output voltage at sudden change of load current in „buck“ mode and at resistive load; $V_{in}=28V$, $V_{out_ref}=14V$; CH1- load current (2A/div), CH2- output voltage (5V/div), TIME: 50ms/div; (a) sudden loading, (b) sudden unloading

Fig.12 (a) shows the responses of the output current (CH1) and the output voltage (CH2) at sudden load. In this case, a transient is observed in the output voltage (voltage drop) of 3V, where its duration was about 5ms. Fig.12(b) shows responses of the output current (CH1) and the output voltage (CH2), but with sudden drop of load current (unloading). In this case, a transient is observed in the output voltage (exceed of voltage of 2.5V), where its duration was about 5ms.

Fig.13 shows the transient voltage responses V_{ds} of the MOSFET synchronous switches M2 and M3, referred to AGND, under the same experimental conditions as in the previous case. Fig. 13(a) shows the voltage responses V_{ds} of the MOSFET switches M2 (channel CH2) and M3 (channel CH1) for the low-load on the output of power converter (at output current of 100mA). It is noticed that the voltage V_{ds} of the switch M2 is equal to 28V (the input voltage of the power converter), while the measured voltage V_{ds} of the switch M3 was 14V (output voltage of the power converter). Fig. 13(b) shows the same responses but in the full load mode of the power converter (at load current 4A).

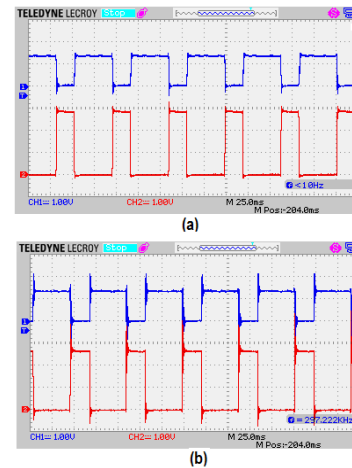


Figure 13. The waveforms of V_{ds} voltage of the MOSFET switch; „buck“ mode; $V_{in}=28V$, $V_{out_ref}=14V$ (resistive load); CH1- voltage V_{ds} of M3(10V/div), CH2- voltage V_{ds} of M2 (10V/div), TIME:3μs/div; (a) low-load, (b) full-load

B) „Boost“ operation (resistive load)

In this mode, characteristic waveforms were recorded, where the input voltage was $V_{in} = 10.8V$. The reference value of the output voltage is set to the value $V_{out_ref} = 14V$ via the "DIP-switch". A 3.5Ω resistive load was connected to the output (the output current was about 4A).

Fig. 14 shows the transient responses of the output voltage during a sudden change in load current. Fig. 14 (a) shows the responses of the output current (CH1) and the output voltage (CH2) at sudden load (changing from 100mA to 4A). In this case, a transient is observed in the output voltage (voltage drop) of 5V, where its duration was about 5ms. Fig.14(b) shows responses of the same magnitudes, but with sudden unloading. In this case, a transient is observed in the output voltage (excess of voltage of 5V), where its duration was about 10ms.

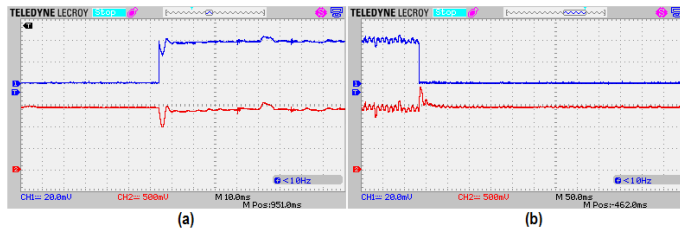


Figure 14. Transient response of output voltage at sudden change of load current in „boost“ mode and at resistive load; $V_{in} = 10.8V$, $V_{out_ref} = 14V$; CH1- load current (2A/div), CH2- output voltage (5V/div); (a) sudden loading, (b) sudden unloading.

Fig.15 shows the transient voltage responses of the V_{ds} MOSFET synchronous switches M2 and M3 referred to AGND, in „boost“ mode.

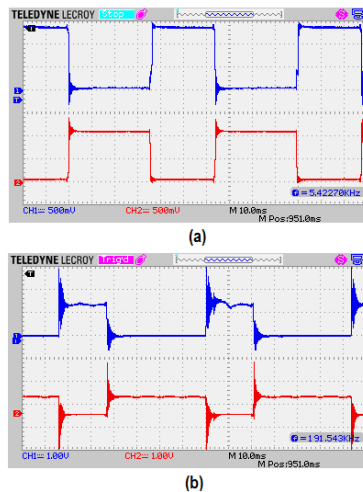


Figure 15. The waveforms of V_{ds} voltage of the MOSFET switch; „boost“ mode; $V_{in} = 10.8V$, $V_{out_ref} = 14V$ (resistive load); CH1- voltage V_{ds} of M3 (5V/div(a) and 10V/div (b)), CH2- voltage V_{ds} of M2 (5V/div(a) and 10V/div(b)), TIME: 3 μ s/div; (a) low-load, (b) full-load.

Fig. 15(a) shows the voltage responses V_{ds} of the MOSFET switches M2 (channel CH2) and M3 (channel CH1) for the

low-load on the output of power converter (at output current of 200mA). It is noticed that the voltage V_{ds} of the switch M2 is equal to 10.8V (the input voltage of the power converter), while the measured voltage V_{ds} of the switch M3 was 14V (output voltage of the power converter). Fig. 15(b) shows the same responses but in the full load mode of the power converter (at load current 4A).

Fig. 16 shows a printed circuit board of the realized "buck-boost four switch power converter".

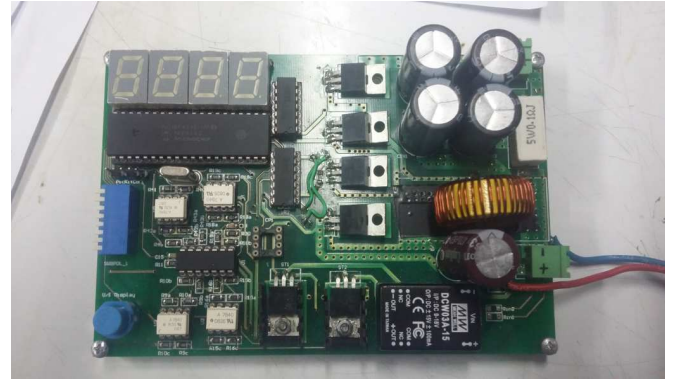


Figure 16. The scope of realized prototype of power converter

VII. CONCLUSIONS

The paper presents the digital topology control of the synchronous DC-DC converter "step-up / step-down four switch converter", which is very interesting for practical applications in "off-grid" solar systems. As a confirmation of the realization of the proposed technical solution, experimental results for the operating modes of the power converter ("buck"/"boost" modes) and for the resistive load are presented. The obtained experimental results are in accordance with the theoretical settings of the converter topology.

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